

AP CSP | Lesson 18: Traversing Lists

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B uses 1-based index 2. A is first; C confuses index and value; D is the length.

Q2. Answer: C

C follows element order. A and B use indices. D sorts without an instruction.

Q3. Answer: D

D initializes once. A and C repeatedly discard progress. B initializes too late.

Q4. Answer: A

A counts three elements. B deduplicates without permission. C confuses values with count. D ignores the elements.

Q5. Answer: B and C

B and C build a stable separate result. A changes the traversed source and violates the design. D adds an unrequested sort; filtering preserves source order here.

Q6. Answer: D

D remains initialized 0 because there are no visits. A assumes indexed access not present. B invents an increment. C refers to a nonexistent element.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 5; (b) 15; (c) 20; (d) 3.

1 mark: After first visit 5.

1 mark: After second visit 15.

1 mark: After third visit 20.

1 mark: Three visits, including both 5 occurrences.

Q8. 4 marks

Answer: After visits 1-4: []; [30]; [30, 30]; [30, 30, 45].

1 mark: After 12: [].

1 mark: After first 30: [30].

1 mark: After second 30: [30, 30].

1 mark: After 45: [30, 30, 45].

Q9. 4 marks

Answer: total $\leftarrow$ 0

FOR EACH minutes IN sessions

```
{
  total  $\leftarrow$  total + minutes
}
DISPLAY(total)
```

1 mark: Initializes total to 0 before the loop.

1 mark: Uses FOR EACH to visit sessions values.

1 mark: Adds current minutes to the running total inside the loop.

1 mark: Displays total after the loop.

Q10. 4 marks

Answer: selected $\leftarrow$ []

FOR EACH minutes IN sessions

```
{
  IF (minutes  $\geq$  goal)
  {
    APPEND(selected, minutes)
  }
}
```

1 mark: Creates empty result before traversal.

1 mark: Traverses sessions without modifying it.

1 mark: Uses inclusive minutes $\geq$ goal condition.

1 mark: Appends each matching current value to selected inside the condition, preserving order.

Q11. 4 marks

Answer: (a) []; (b) [30]; (c) [31]; (d) [].

1 mark: Below threshold: [].

1 mark: At threshold: [30].

1 mark: Above threshold: [31].

1 mark: Empty input: [].

Q12. 4 marks

Answer: (a) 2. (b) Initialize total once before traversal. (c) 13. (d) Keeping the source unchanged prevents shifts from altering which elements are visited and preserves data for checking.

1 mark: Buggy final total 2.

1 mark: Moves initialization before the loop.

1 mark: Correct full sum 13.

1 mark: Explains stable visits or preserving the original data as a concrete reason.